Calculate required GPA to Target

```
DELIMITER $$

CREATE PROCEDURE CalculateTargetGPA(
   IN p_student_id INT,
   IN p_target_gpa DECIMAL(3,2),
   IN p_total_program_credits INT
)
BEGIN
   DECLARE completed_credits INT;
   DECLARE current_points DECIMAL(10,2);
   DECLARE required_points DECIMAL(10,2);
   DECLARE remaining_credits INT;
   DECLARE required_gpa DECIMAL(3,2);

   Get completed credits and current grade points
   SELECT SUM(c.credit_hours),
        SUM(g.grade_point * c.credit_hours)
   INTO completed_credits, current_points
   FROM grades g
   JOIN courses c ON g.course_id = c.course_id
   WHERE g.student_id = p_student_id;

   Calculate remaining credits
   SET remaining_credits = p_total_program_credits - completed_credits;

   Calculate required total points
   SET required_points = p_target_gpa * p_total_program_credits;

   Calculate GPA needed in remaining courses
   IF remaining_credits > 0 THEN
      SET required_gpa = (required_points - current_points) / remaining_credits;
   ELSE
      SET required_gpa = 0;
   END IF;

   Output result
   SELECT required_gpa AS required_gpa_needed;

END $$

DELIMITER ;
```

Calculate and store gpa

```
DELIMITER $$

CREATE PROCEDURE CalculateGPA(IN p_student_id INT)
BEGIN
   DECLARE total_points DECIMAL(10,2);
   DECLARE total_credits INT;
   DECLARE calculated_gpa DECIMAL(3,2);

   Calculate total grade points
   SELECT SUM(g.grade_point * c.credit_hours),
        SUM(c.credit_hours)
   INTO total_points, total_credits
   FROM grades g
   JOIN courses c ON g.course_id = c.course_id
   WHERE g.student_id = p_student_id;

   Calculate GPA
   IF total_credits > 0 THEN
      SET calculated_gpa = total_points / total_credits;
   ELSE
      SET calculated_gpa = 0;
   END IF;

   Update GPA table
   UPDATE gpa
   SET gpa_value = calculated_gpa
   WHERE student_id = p_student_id;

END $$

DELIMITER ;
```